Software Test Plan

Web-Based ML Model Evaluator

Project	Web-Based ML Model Evaluator
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Related documents	SRS ,SAD, RTM

Revision History

Version	Date	Author	Change summary
0.2	12-10-2026	Member 1,2,3,4	Integrated sections; aligned requirement IDs and security traceability

1. Introduction

Purpose: This document defines the test plan for the Web-Based ML Model Evaluator (WBMME) v1.0. It describes the objectives, scope, strategy, resources, schedule and responsibilities for testing, and lists the test cases that verify the requirements in the SRS.

Scope: Testing covers user registration and login, role-based access, CSV dataset upload and preparation, model selection and training, evaluation metrics and charts, experiment history and comparison, report export, administration, and the stated performance, portability, usability and security requirements. Internals of third-party libraries and the hosting platform are excluded.

References: WBMME SRS v1.0; WBMME SAD v1.0 (Sections 3.3 to 3.9); SE Mini-Project Deliverables Part-1 and Part-2; Jackfruit SE Mini-Project Guidelines 2026; IEEE 829 (test documentation); OWASP Top 10.

Definitions: WBMME (Web-Based ML Model Evaluator), SRS (Software Requirements Specification), SAD (Software Architecture Document), RTM (Requirements Traceability Matrix), TC (Test Case), CSV (Comma-Separated Values), UAT (User Acceptance Testing), CSRF (Cross-Site Request Forgery), STRIDE (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege).

2. Test Items

- Authentication and access control module (registration, login, roles)
- Data workspace module (CSV upload, validation, profiling, preprocessing, train/test split)
- Model selection and training module
- Evaluation module (metrics, confusion matrix, feature importance)
- Experiment management and reporting module (history, comparison, export)
- Admin interface (user and dataset management)
- REST/JSON endpoints: /api/upload, /api/train, /api/evaluate, /api/runs
- Logging, monitoring and input validation controls

3. Features to be Tested

Features are listed with the requirement IDs of the SRS (Part B, sections 4 and 5). “Covered by” names the test case (Appendix A) or the security check (section 5.1) that verifies the requirement; “planned” means that it is scheduled in the sprint shown and has no test case among the ten in Appendix A yet.

3.1 Functional features

Req ID	Feature	Priority	Covered by
MLE-F-001	User registration and login	High	TC-AUTH-01
MLE-F-002	Role-based access (Data Scientist, Student, Admin)	High	TC-SEC-01
MLE-F-003	Upload CSV dataset with validation	High	TC-DATA-01, TC-DATA-02
MLE-F-004	Preview and profile the dataset	High	TC-DATA-01
MLE-F-005	Handle missing values and encode categories	High	Planned (Sprint 1)
MLE-F-006	Configure target column and train/test split	High	TC-MODEL-01
MLE-F-007	Select a model from the catalogue	High	TC-MODEL-01
MLE-F-008	Configure hyperparameters	Medium	TC-MODEL-01
MLE-F-009	Train a model and show status	High	TC-MODEL-01
MLE-F-010	Compute accuracy, precision, recall, F1	High	TC-EVAL-01
MLE-F-011	Show confusion matrix and plots	High	TC-EVAL-01
MLE-F-012	Feature importance / explanation	Medium	Planned (Sprint 2)
MLE-F-013	Save each run (history)	High	TC-EXP-01
MLE-F-014	Compare runs side by side	Medium	TC-EXP-01

Req ID	Feature	Priority	Covered by
MLE-F-015	Export results report	Medium	TC-DATA-02 (step 3) (planned, Sprint 2)
MLE-F-016	Admin manages users and datasets	High	TC-SEC-01

3.2 Non-functional features

Req ID	Feature	Priority	Covered by
MLE-NF-001	Upload of up to 10 MB processed in 5 s or less	High	TC-PERF-01
MLE-NF-002	Training of standard models on up to 50,000 rows in 60 s or less	High	TC-PERF-02
MLE-NF-003	Pages load in 3 s or less for 20 concurrent users	Medium	TC-PERF-03
MLE-NF-004	Runs on Windows, Linux and macOS via Docker	Medium	Planned (Sprint 2)
MLE-NF-005	Usable by a first-time user without training	Medium	Planned (Sprint 2)
MLE-NF-006 (new)	Failed jobs are contained; saved data survives restart	Medium	Planned (Sprint 2)
MLE-NF-007 (new)	Keeps page-load limits with 20 datasets and 100 runs per user	Low	Planned (Sprint 2)

3.3 Security features

Req ID	Feature	Priority	Covered by
MLE-SR-001	Passwords stored hashed	High	TC-AUTH-01, SV-01
MLE-SR-002	All traffic over HTTPS (TLS 1.2+)	High	SV-02 (planned, Sprint 2)
MLE-SR-003	Uploads validated for type, size and content	High	TC-DATA-02, SV-03
MLE-SR-004	Only system-created model artifacts are loaded	High	SV-04 (planned, Sprint 2)
MLE-SR-005	Per-user data isolation and audit log of actions	High	TC-SEC-01, SV-05
MLE-SR-006 (new)	Request, job and run-time limits against abuse	High	SV-06 (planned, Sprint 2)
MLE-SR-007 (new)	Session expiry, login throttling and CSRF protection	High	TC-AUTH-01 (cookie flag only), SV-07, SV-08 (planned, Sprint 2)

4. Features not to be Tested

Item	Reason
Internal correctness of third-party libraries (Django, pandas, scikit-learn, joblib, Chart.js)	Assumed tested by their maintainers. WBMME is tested for how it uses them, for example by comparing metrics with a scikit-learn reference (TC-EVAL-01).
Predictive quality or production suitability of the models a user trains	The system only trains, evaluates and reports. It makes no claim about model quality (SRS 1.2).
Browser internals and operating-system behaviour	Out of the project's control. Only current Chrome and Firefox on Windows, Linux and macOS are supported (MLE-NF-004).
Mobile and tablet layouts	Outside the SRS operating environment (SRS 2.4).
SHAP explanations, PDF export and Celery job queues	Optional, not committed in the SRS (MLE-F-012, MLE-F-015, SAD 3.6). They are tested only if implemented.
Load beyond 20 concurrent users; production-scale deployment tuning	The SRS commits to 20 concurrent users (MLE-NF-003) and a Docker deployment only.
Penetration testing beyond the OWASP ZAP baseline scan and the abuse cases of section 5.1	Out of scope for a two-sprint student project; the checks in 5.1 cover each security requirement.
Hosting platform and network infrastructure	Not supplied by the project. TLS is checked on the deployed build only (SV-02).

5. Test Approach / Strategy

Levels

Level	Scope	Tools
Unit	Module-level tests of services: validation, preprocessing, training, metrics, access checks. Coverage target 70% or higher for service modules (section 14).	pytest, pytest-cov
Integration	API layer to services, services to database and file store, and the pipeline stages in sequence (Upload, Preprocess, Train, Evaluate, Record).	pytest, Django test client, Postman
System	End-to-end flows through the browser: login, upload, split, train, evaluate, history, compare, export, administration.	Selenium, manual execution
Acceptance (UAT)	Walkthrough of the demo scenarios AC-1 to AC-5 (SRS 6.2) by the team and the faculty reviewer.	Manual

Types

- **Functional testing** of every feature in section 3, using the test cases of Appendix A.

- **Regression testing** on every pull request, automated in GitHub Actions.
- **Performance testing** against MLE-NF-001, NF-002 and NF-003 on one fixed machine, with its specification recorded.
- **Usability testing** with five first-time users (MLE-NF-005).
- **Reliability testing** by fault injection (invalid data, time-out) and a server restart (MLE-NF-006).
- **Portability checks** by building and starting the Docker image on Windows, Linux and macOS (MLE-NF-004).
- **Security testing** as described in section 5.1.

Test design techniques

- **Boundary value analysis and equivalence partitioning** for every limit: file size (9.9, 10 and 10.1 MB), rows (19, 20, 50,000 and 50,001), parameter ranges (for example C = 0.01, 100, 0 and 101), test share (10% and 50%).
- **Valid and invalid data sets** for every input, as the course test-case template requires.
- **Oracle comparison:** metrics are compared with a scikit-learn reference script that uses the same split and seed (tolerance 0.001).
- **Error guessing and abuse cases** derived from the STRIDE table in SAD 3.9.

Defect severity

Severity	Meaning
Critical	Data loss, authentication bypass, or data exposed between users. Testing is suspended (section 12).
High	A committed requirement fails with no workaround, or a security check fails.
Medium	A requirement fails but a workaround exists, or a non-committed feature fails.
Low	Cosmetic or wording issues.

Entry criteria: A stable build is merged to the main branch, the test data and the test environment are ready, and the relevant SRS requirements are baselined.

Exit criteria: 100% of the planned test cases for the sprint are executed, no critical or high-severity defect remains open, all acceptance criteria (SRS 6.2) are satisfied, and deviations from the SRS, SAD and this plan are documented for the final demo.

5.1 Security Validation

Security validation verifies the three security objectives of SRS 5.1.1, which match the objectives of SAD 3.9: confidentiality (MLE-SO-001), integrity (MLE-SO-002) and availability (MLE-SO-003).

Each check below verifies one security requirement and is traceable to it. The checks run on the deployed Docker build with HTTPS, which is available before Sprint 2 (section 11).

Table 5.1-1: Security objectives and their checks

Objective	Requirements	Checks	Threats (STRIDE)
Confidentiality MLE-SO-001	MLE-SR-001, SR-002, SR-005(a), SR-007; MLE-F-002	SV-01, SV-02, SV-05, SV-07, SV-08	Spoofing, information disclosure, elevation of privilege
Integrity MLE-SO-002	MLE-SR-003, SR-004, SR-005(b)	SV-03, SV-04, SV-05	Tampering, repudiation
Availability MLE-SO-003	MLE-SR-006	SV-06	Denial of service
All objectives	MLE-SR-001 to SR-007	SV-09, SV-10	All

Confidentiality checks

ID	Req.	What is verified	Method and tool	Pass criterion
SV-01	SR-001	Passwords are stored only as salted PBKDF2 or bcrypt hashes; no password appears in logs or responses.	Register a user, inspect the user table and the application logs; extend TC-AUTH-01 step 5 with a grep of the logs.	Stored value starts with a PBKDF2 or bcrypt identifier; the password text is found nowhere.
SV-02	SR-002	Only TLS 1.2 and above is accepted; HTTP is redirected or refused.	Run testssl.sh (or nmap ssl-enum-ciphers) and curl against the deployed HTTPS build.	TLS 1.0 and 1.1 refused; http:// redirects to https://; no weak cipher reported.
SV-05	SR-005	Users cannot read or change other users' data; every listed action is written to the audit log without raw data or credentials.	Change dataset and run IDs in URLs and API calls as another user; open admin pages as a Student; review the audit log (TC-SEC-01).	403 or 404 and no data returned; log shows the denied attempts with user ID and timestamp.
SV-07	SR-007	Sessions expire, cookies carry the HttpOnly and Secure flags, repeated failed logins are blocked.	Log in, then reuse the session after the (shortened) inactivity limit; inspect cookies in the browser tools; submit 5 wrong passwords in 10 minutes and then the right one.	Expired session is redirected to login; flags present; sixth attempt is blocked for 15 minutes even with the right password.
SV-08	SR-007	State-changing requests without a valid CSRF token are refused.	Replay a POST (upload, train, delete) from Postman without the token and with a wrong token.	HTTP 403 and no state change.

Integrity checks

ID	Req.	What is verified	Method and tool	Pass criterion
SV-03	SR-003	Upload validation rejects wrong type, wrong MIME type, oversized, empty and malformed files; formula cells are neutralised in exports.	Upload the invalid files of TC-DATA-02, then fuzz the upload with truncated, binary and very wide CSV files (Postman or a Python script).	Every invalid file is rejected with a clear message and nothing is stored; no unhandled exception.
SV-04	SR-004	Uploaded model files are refused, only system-created artifacts are loaded, and artifacts cannot be fetched by URL.	Send a .pkl and a .joblib file to /api/upload and any other endpoint; request an artifact path directly; search the code for joblib.load and pickle.load calls.	Model files refused; direct artifact URL returns 404; every load call uses a path created by the system.

Availability checks

ID	Req.	What is verified	Method and tool	Pass criterion
SV-06	SR-006	Request size, request rate, job quota and job time-out limits are enforced.	Send an 11 MB body; send 130 requests in one minute as one user; start a third job as a Data Scientist and a second job as a Student; run a job that exceeds 300 s (lower the limit in the test settings to shorten the run).	HTTP 413; HTTP 429 after the 120th request; extra jobs refused with a message; long job ends as Failed with reason time-out; the server answers other users throughout.

Checks across all objectives

ID	Req.	What is verified	Method and tool	Pass criterion
SV-09	All SR	Automated scan finds no high-severity web vulnerability.	OWASP ZAP baseline scan against the Docker build, run in the CI pipeline or manually before each sprint demo.	No High alerts open; Medium alerts reviewed and recorded.
SV-10	All SR	Static analysis finds no unresolved high-severity security issue in the code.	Run SonarQube (or bandit and flake8) in GitHub Actions on every pull request.	No open blocker or high-severity security finding at merge.

Tools: OWASP ZAP baseline scan, Postman negative tests, testssl.sh or nmap for TLS, SonarQube or bandit in GitHub Actions, and manual abuse cases derived from the STRIDE table in SAD 3.9. A failed security check is a high-severity defect; a failed check that exposes data between users or bypasses authentication is critical.

6. Test Environment

Hardware: Developer laptops (minimum 8 GB RAM, multi-core CPU) for functional tests; one fixed machine for performance tests so results are comparable.

Software: Python 3.11, Django 5, pandas, scikit-learn, joblib, SQLite (development) and optionally PostgreSQL, Docker; browsers: current Chrome and Firefox; operating systems: Windows, Linux, macOS.

Tools: pytest and pytest-cov (unit and integration), Selenium (UI automation), Postman (API), JMeter or Locust (performance), OWASP ZAP (security scan), GitHub Actions (CI), GitHub Issues and Projects (defect tracking).

Test data: Small sample datasets (for example Iris-like and Titanic-like CSV files), a generated 10 MB CSV, a generated 50,000-row CSV, and invalid files (wrong type, empty, oversized, malformed, formula-injection cells). Dummy accounts for each role: Data Scientist, Student and Admin.

7. Test Schedule

The schedule follows the sprint plan in Mini-Project Deliverables Part-2.

Milestone	Dates	Owner
Test case design and review	10-Oct-2026 to 12-Oct-2026	Members 2 and 4
Environment setup, CI pipeline, sprint dry run	12-Oct-2026 to 16-Oct-2026	Members 1 and 3
Sprint 1 test execution	19-Oct-2026 to 23-Oct-2026	All members
Sprint 2 test execution and regression	26-Oct-2026 to 30-Oct-2026	All members
Final UAT and Test Summary Report	29-Oct-2026 to 30-Oct-2026	Member 1

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual and automated), Section 13 and Appendix A
- Test scripts (pytest, Selenium, Postman collection, JMeter or Locust plan)
- Test data sets
- Test execution logs and CI run reports
- Defect reports (GitHub Issues)
- Sprint 1 and Sprint 2 demo videos
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Test Lead and document owner	Member 1	Introduction, test items, environment, schedule and deliverables; merges and formats the final plan
Test design lead	Member 2	Features, strategy, security validation and SRS traceability

Role	Name	Responsibility
Risk and quality lead	Member 3	Risks, assumptions, suspension criteria, metrics and approvals
Test case author	Member 4	Writes and maintains the 10 test cases and updates the RTM with Member 2
Repo owner / reviewer	Team repo owner	Reviews pull requests and confirms tests pass before merge
Faculty guide	Course faculty	Reviews and signs off test results and deviations

13. Test Case Management and Traceability

The Requirements Traceability Matrix (RTM) maps every SRS requirement to the test case or security check that verifies it. The full test cases are in Appendix A (seven functional and three non-functional). The SRS RTM (SRS section 8) shows the same mapping from the requirement side and also traces each requirement to its use case, architecture component and design reference.

Req ID	Requirement (short)	Test case(s) / check	Level	Status
MLE-F-001	User registration and login	TC-AUTH-01	System	Defined
MLE-F-002	Role-based access (Data Scientist, Student, Admin)	TC-SEC-01	System	Defined
MLE-F-003	Upload CSV dataset with validation	TC-DATA-01, TC-DATA-02	Integration	Defined
MLE-F-004	Preview and profile the dataset	TC-DATA-01	Integration	Defined
MLE-F-005	Handle missing values and encode categories	Planned (Sprint 1)	Unit, integration	Planned
MLE-F-006	Configure target column and train/test split	TC-MODEL-01	System	Defined
MLE-F-007	Select a model from the catalogue	TC-MODEL-01	System	Defined
MLE-F-008	Configure hyperparameters	TC-MODEL-01	System	Defined
MLE-F-009	Train a model and show status	TC-MODEL-01	System	Defined
MLE-F-010	Compute accuracy, precision, recall, F1	TC-EVAL-01	System	Defined
MLE-F-011	Show confusion matrix and plots	TC-EVAL-01	System	Defined
MLE-F-012	Feature importance / explanation	Planned (Sprint 2)	Unit, system	Planned
MLE-F-013	Save each run (history)	TC-EXP-01	System	Defined
MLE-F-014	Compare runs side by side	TC-EXP-01	System	Defined
MLE-F-015	Export results report	TC-DATA-02 (step 3) (planned, Sprint 2)	System	Partly defined
MLE-F-016	Admin manages users and datasets	TC-SEC-01	System	Defined

Req ID	Requirement (short)	Test case(s) / check	Level	Status
MLE-NF-001	Upload of up to 10 MB processed in 5 s or less	TC-PERF-01	Performance	Defined
MLE-NF-002	Training of standard models on up to 50,000 rows in 60 s or less	TC-PERF-02	Performance	Defined
MLE-NF-003	Pages load in 3 s or less for 20 concurrent users	TC-PERF-03	Performance	Defined
MLE-NF-004	Runs on Windows, Linux and macOS via Docker	Planned (Sprint 2)	System	Planned
MLE-NF-005	Usable by a first-time user without training	Planned (Sprint 2)	Usability	Planned
MLE-NF-006 (new)	Failed jobs are contained; saved data survives restart	Planned (Sprint 2)	System	Planned
MLE-NF-007 (new)	Keeps page-load limits with 20 datasets and 100 runs per user	Planned (Sprint 2)	Performance	Planned
MLE-SR-001	Passwords stored hashed	TC-AUTH-01, SV-01	System	Defined
MLE-SR-002	All traffic over HTTPS (TLS 1.2+)	SV-02 (planned, Sprint 2)	Security	Planned
MLE-SR-003	Uploads validated for type, size and content	TC-DATA-02, SV-03	Integration	Defined
MLE-SR-004	Only system-created model artifacts are loaded	SV-04 (planned, Sprint 2)	Security	Planned
MLE-SR-005	Per-user data isolation and audit log of actions	TC-SEC-01, SV-05	System	Defined
MLE-SR-006 (new)	Request, job and run-time limits against abuse	SV-06 (planned, Sprint 2)	Security	Planned
MLE-SR-007 (new)	Session expiry, login throttling and CSRF protection	TC-AUTH-01 (cookie flag only), SV-07, SV-08 (planned, Sprint 2)	Security	Partly defined

Coverage summary

Requirement group	Total	Fully covered	Open items
Functional (MLE-F)	16	13	F-005, F-012, F-015
Non-functional (MLE-NF)	7	3	NF-004, NF-005, NF-006, NF-007
Security (MLE-SR)	7	3	SR-002, SR-004, SR-006, SR-007

“Fully covered” means a formal test case of Appendix A exists. Every security requirement also has a security check (SV-nn) in section 5.1. The open functional and non-functional items are

scheduled in Sprint 1 or Sprint 2 (section 7); Member 4 can close them by adding test cases to Appendix A, and this table is then updated.